

MARWADI UNIVERSITY

DEPARTMENT OF AI, ML

DAA (01AI0506) - Lab Practical

Practical 04: Implementation and Time Analysis of Factorial using Iterative and Recursive Methods

Name: K.Raju Enrollment No: 92460118432

Aim

Implementation and Time Analysis of Factorial using Iterative and Recursive Methods as specified in the DAA Lab Manual.

Program

```
#include <iostream>
using namespace std;

long long factorialIterative(int n) {
    long long fact = 1;
    for (int i = 2; i <= n; i++) fact *= i;
    return fact;
}

long long factorialRecursive(int n) {
    if (n <= 1) return 1;
    return n * factorialRecursive(n - 1);
}

int main() {
    int n = 5;
    cout << "Iterative: " << factorialIterative(n) << endl;
    cout << "Recursive: " << factorialRecursive(n) << endl;
}
```

Output

```
Iterative: 120
Recursive: 120
```

Time and Space Analysis

Iterative: Time $\Theta(n)$, Space $\Theta(1)$. Recursive: Time $\Theta(n)$, Space $\Theta(n)$. Recurrence: $T(n) = T(n-1) + c$.

Result

The practical was implemented successfully and the expected output was obtained.